Automated Web Scanner

Project README — IBM SkillsBuild Cyber Security 2026

A beginner-friendly Python cyber security project that performs a basic, passive check of HTTP security headers and CORS-related response headers.

Features

• Checks common security headers: HSTS, CSP, X-Content-Type-Options, X-Frame-Options, Referrer-Policy, and Permissions-Policy.

• Displays HTTP status and whether the final URL uses HTTPS.

• Reports basic CORS response settings.

• Can save results to a JSON file.

• Uses one normal GET request; it does not exploit vulnerabilities or attempt authentication bypass.

Requirements

Python 3.9 or newer and the requests library.

Setup

```
python -m venv .venv
```

Activate the environment:

```
Windows: .venv\Scripts\activate
macOS/Linux: source .venv/bin/activate
```

Install dependencies:

```
pip install -r requirements.txt
```

Run

Only scan websites you own or have explicit permission to assess.

```
python automated_web_scanner.py https://example.com
```

Save a JSON report:

```
python automated_web_scanner.py https://example.com --json report.json
```

Jupyter Notebook

Open YourName_AutomatedWebScanner.ipynb in Jupyter Notebook or Google Colab and run the cells in order. Replace the example URL with a site you are authorized to test.

Understanding the Results

• A missing header is a review item, not automatic proof that a site is vulnerable.

• CORS behavior depends on request context and browser enforcement.

• Review results against the website's intended security policy.

Limitations

• This is an educational first-pass scanner, not a full penetration-testing tool.

• It does not crawl pages, test authenticated routes, or prove exploitability.

• Network errors, redirects, proxies, and server-specific behavior can affect results.

Project Files

• automated_web_scanner.py — command-line scanner

• YourName_AutomatedWebScanner.ipynb — notebook version

• requirements.txt — dependency list

• YourName_ProjectReport.docx — project report

• README.md — project overview and instructions

Ethical Use

Run the scanner only against websites you own or for which you have explicit permission. This tool provides a basic configuration review and is not a substitute for a complete security audit.